

Integrating Neural Networks into Resource-Constrained IoT Devices: Challenges and Techniques

Muhammad Saad Azam
Department of Computer Science
University of Engineering and Technology
Lahore, Pakistan

Abstract—Abstract

Index Terms—component, formatting, style, styling, inserts

I. INTRODUCTION

Traditional statistical machine learning models have been an integral part of IoT device infrastructure. These models are generally memory-efficient, computationally inexpensive, and capable of achieving fairly accurate predictions in many practical scenarios. Due to these characteristics, they have historically been preferred for deployment in resource-constrained IoT environments. Among these statistical models, Random Forest classifiers and similar decision forest methods have consistently performed well for a variety of IoT tasks. These models are particularly effective in scenarios where the available datasets are relatively small, as they tend to remain robust even when training data is limited [1], [2]. When the underlying data is simple, well-structured, and easily interpretable, such models are often more than sufficient for achieving reliable performance in IoT applications.

However, these traditional machine learning models often struggle to learn complex representations when the underlying data itself is highly complicated. Existing literature notes that although decision forests are memory-efficient and practical for constrained environments, they remain limited in expressiveness. As a result, they frequently require dynamic adaptations, such as pruning strategies or meta-learning mechanisms, in order to handle more complex patterns and concept drift in data streams [3]. Similar limitations can be observed in other statistical machine learning models that rely heavily on handcrafted features and relatively shallow model structures. Furthermore, when the input data becomes high-dimensional or unstructured—such as visual, auditory, or multimodal data—these statistical models often fail to scale effectively, which significantly restricts their ability to capture deeper patterns within the data [4].

This is precisely the domain in which neural networks excel. Neural networks (NNs) are capable of learning complex hierarchical representations directly from raw data, enabling them to model intricate relationships that traditional statistical models often fail to capture. In addition, neural networks demonstrate strong scalability with respect to both the number of model parameters and the size of the datasets used during training [5].

Their representational power is further enhanced through the use of activation functions that introduce non-linearity into the learning process. Moreover, various architectural innovations like Convolutional Neural Networks (CNNs) for visual data [6] and Transformer architectures for sequence modeling [7] have significantly expanded the ability of neural networks to learn, generalize, and perform effectively across a wide range of complex tasks. As a result, neural networks have become a dominant approach for processing high-dimensional and unstructured data generated by modern IoT systems.

However, despite their potential, there are certain challenges that are prominent with neural networks. Efficient NNs are usually large [8] and therefore cannot be efficiently integrated to lightweight, hardware-restricted IoT devices. IoT devices typically have limited memory, storage and battery life. They also lack powerful processors and GPUs. Furthermore, raw data transfer between cloud, routers and IoT devices can be expensive as well. This greatly discourages integration of NNs in IoT devices and reduces their practical utility. As a result, substantial research efforts have been made to effectively integrate NNs in lightweight IoT devices.

Numerous developments have been made for lightweight integration of deep learning in resource-constrained environments, targetting different aspects of the deep learning ecosystem like hardware, network architecture, deployments and more [9]. However, memory, performance and hardware constraints of IoT devices make the integration even more challenging and more harmonized and considerable research is required in this aspect. This survey attempts to fill that gap.

II. IOT LANDSCAPE AND CONSTRAINTS

A. Device Tiers

Based on the tasks and their capabilities, we are going to divide IoT devices into three tiers. Some of them are more capable than the others. Each have their own place in the overall IoT architecture.

1) *Tier 1: Ultra-constrained Microcontrollers:* These IoT devices have the smallest SRAM and flash memory. The SRAM typically ranges from 16 to 512kb while flash storage is typically less than 2MB. The power consumption is less than 1mW in active mode and have no operating systems. Device families like ARM Cortex M0/M0+, ultra low-power

NPU and RISC-V ultra-constrained MCUs come under this tier. The TinyML framework primarily targets devices under this tier [10].

2) *Tier 2: Capable Processors*: Devices like Raspberry Pi, Nvidia Jetson Nano, and ARM Cortex-A based SOCs can be categorized under this tier. The RAM ranges from hundred megabytes to a few gigabytes with optional hardware accelerators. These devices allow us to deploy and infer deep learning models that are more accurate and efficient. Such devices can run minimal operating systems like RIOT OS [11]. Models like MobileNet and ResNet can be used for inference on these devices.

3) *Tier 3: Edge Servers and Gateways*: Devices within this tier represent the most capable class of hardware, typically offering gigabytes of memory, substantial storage capacity, and highly efficient high-performance processors. These platforms are capable of running full-scale deep learning models, including convolutional neural networks (CNNs) and transformer-based architectures. Commonly deployed in edge AI systems and cloud-connected infrastructures, they often serve as aggregation points for data collected from lower-tier IoT devices. They are also able to run operating system like Linux. As these devices are already fairly capable of running heavy models, these devices are not a primary focus of this survey.

In a typical IoT architecture, all these devices intercommunicate with each other. Among the hardware tiers discussed, ultra-constrained microcontrollers present the greatest challenges for deep learning deployment because of their severe limitations in memory, storage, processing capability, and energy availability. As a result, much of the recent research in TinyML, embedded AI, and resource-aware model optimization has focused specifically on enabling intelligent inference on these highly constrained devices.

B. Key Constraints

Here are some key constraints faced by IoT devices:

1) *Memory*: SRAM is the dominant bottleneck. Ultra-constrained MCUs rely solely on SRAM for computation. Standard deep learning models generate activation tensors that can easily exceed SRAM capacity.

2) *Computation*: MCUs (Microcontroller Units) usually have a clock cycle of 48MHz to 480MHz and limited instruction parallelism. This forces our hand to carry out MAC (multiply-accumulate) more efficiently within limited constraints. Architectures like SqueezeNet [12] and MobileNet [18] can reduce the number of operations, making such compressed architectures more suitable for IoT devices.

3) *Energy*: Most IoT deployments utilize battery-powered or energy-harvesting devices that impose strict power budget. Energy consumption during inference and training depends on the memory, numerical precision and arithmetic computation. Furthermore, DRAM consumes way more energy per bit compared to SRAM.

4) *Numerical Precision*: Typically, models use 32-bit precision floating points but that consumes a lot of SRAM. To fit such models within microcontrollers, some kind of

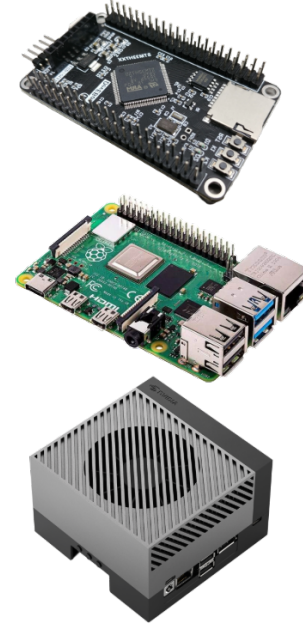


Fig. 1. Device of each tier. 1st one is STM32, 2nd Raspberry PI 4, and 3rd is Nvidia Jetson AGX Orin

compression is required. Quantization and pruning can help us out here, with a trade-off being the accuracy of those models.

5) *Storage*: The weights of the model reside in a flash memory, where a typical flash memory consists of 512kb to 8MB. Models like ResNet-50 [14] requires around 98MB of storage with float32 precision. Quantization to 4-bit compresses the size to 12MB which is still a lot for MCUs. Hence, pruning and low-bit quantization is an important requirement for compression.

6) *Latency*: Most of the IoT devices for anomaly, object and gesture detection require a response of milliseconds. Hence, the models deployed should meet the real-time requirements of these IoT devices, with quick inference and minimal latency.

III. IMPLEMENTATION STRATEGIES

Based on these constraints, the implementation, optimization, and deployment of deep learning models can be approached from multiple directions. The key areas of focus include neural network architectures, model compression strategies, hardware advancements and their suitability for AI integration, as well as lightweight frameworks that enable efficient deployment of AI on MCUs.

In this survey, we analyze these strategies in terms of their computational complexity and suitability for MCU deployment. We further examine the trade-offs between computation, memory usage, performance, and accuracy to identify practical operating points and highlight any standout approaches. Existing benchmarks are catalogued based on their computational requirements, performance, and accuracy trade-offs, with the analysis conducted primarily through the lens of IoT devices.

IV. NETWORK ARCHITECTURE

Our first area of interest in network architecture, where various neural networks of different compositions are introduced. Some architectures excel in accuracy while others chiefly focus on performance with minimal parameters. When it comes to vision tasks, CNNs are fairly competent on lightweight IoT devices, providing a favorable balance between performance and accuracy. However, since the introduction of transformer architecture [7], vision transformers have gathered a lot of attention and growing amount of research is being done to make them more lightweight and efficient. First, we outline models utilizing CNN architecture, their trade-offs and then the vision transformers.

A. Lightweight CNNs

1) *SqueezeNet*: SqueezeNet was introduced in 2016 with the goal of matching AlexNet's [15] performance with greatly reduced parameter count with minimal memory footprint [12]. SqueezeNet achieves this reduction by using 1×1 convolution filters instead of 3×3 convolutions. The squeeze layers reduce the number of input channels to subsequent filters. The resulting model has a memory footprint of approximately 4.8MB. This model can be further compressed with deep compression [16] (pruning, quantization to 6-bit and Huffman coding), resulting in a light 0.48MB model. This deep compression achieved comparable performance to the uncompressed one. While uncompressed SqueezeNet is not suitable for MCUs, there have been deployments of SqueezeNet-like CNNs on microcontrollers like Arduino Portenta H7 Lite [17], making SqueezeNet-inspired architectures a potential candidate for tier 1 devices.

2) *MobileNet*: The MobileNet series of models is a lightweight architecture of Neural Networks that achieved fairly high accuracy by reducing the computation at inference and lowering the number of parameters [18]. The core innovation of this architecture came from separation of spatial filtering and channel mixing using small filters, which resulted in high accuracy with reduced computation and size. Comparing in to SqueezeNet, the amount of FLOPs (Floating Point Operations) is greatly reduced in MobileNet which makes it even more suitable for lightweight devices with deep compression, as it has been deployed to MCUs like STM32H7 [20]. The latest release of MobileNetV4 further enhanced the accuracy by utilizing Universal Inverted Bottleneck (UIB) and Mobile Multi-Query Attention (MQA) [19]

REFERENCES

- [1] B. Sliwa, N. Piatkowski and C. Wietfeld, "LIMITS: Lightweight Machine Learning for IoT Systems with Resource Limitations," ICC 2020 - 2020 IEEE International Conference on Communications (ICC), Dublin, Ireland, 2020, pp. 1-7, doi: 10.1109/ICC40277.2020.9149180.
- [2] Y. Meidan, M. Bohadana, A. Shabtai, M. Ochoa, N. O. Tippenhauer, J. D. Guarnizo, and Y. Elovici, "Detection of Unauthorized IoT Devices Using Machine Learning Techniques," arXiv:1709.04647, Sep. 2017.
- [3] A. Lourenço, J. Rodrigo, J. Gama, and G. Marreiros, "On-device edge learning for IoT data streams: a survey," arXiv:2502.17788, Feb. 2025.
- [4] S. Zhou et al., "A Comprehensive Survey on Deep Clustering: Taxonomy, Challenges, and Future Directions," ACM Computing Surveys, vol. 56, no. 7, pp. 1-56, Mar. 2024. DOI: 10.1145/3617813
- [5] J. Hestness, S. Narang, N. Ardalani, G. Diamos, H. Jun, H. Kianinejad, M. M. A. Patwary, Y. Yang, and Y. Zhou, "Deep Learning Scaling is Predictable, Empirically," arXiv:1712.00409, Dec. 2017.
- [6] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," Proceedings of the IEEE, vol. 86, no. 11, pp. 2278-2324, Nov. 1998.
- [7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in Advances in Neural Information Processing Systems 30 (NIPS 2017), Long Beach, CA, USA, Dec. 2017, pp. 6000-6010.
- [8] J. Kaplan et al., "Scaling Laws for Neural Language Models," arXiv:2001.08361, Jan. 2020.
- [9] H.-I. Liu, M. Galindo, H. Xie, L.-K. Wong, H.-H. Shuai, Y.-H. Li, and W.-H. Cheng, "Lightweight deep learning for resource-constrained environments: A survey," ACM Computing Surveys, vol. 56, no. 7, pp. 1-42, Jul. 2024. doi: 10.1145/3657282.
- [10] J. Lin, L. Zhu, W. -M. Chen, W. -C. Wang and S. Han, "Tiny Machine Learning: Progress and Futures [Feature]," in IEEE Circuits and Systems Magazine, vol. 23, no. 3, pp. 8-34, thirdquarter 2023, doi: 10.1109/MCAS.2023.3302182.
- [11] E. Baccelli, C. Gündoğan, O. Hahm, P. Kietzmann, M. Lenders, H. Petersen, K. Schleiser, T. C. Schmidt, and M. Wählich, "RIOT: An open source operating system for low-end embedded devices in the IoT," IEEE Internet of Things Journal, vol. 5, no. 6, pp. 4428-4440, Dec. 2018. doi: 10.1109/IIOT.2018.2815038.
- [12] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, and K. Keutzer, "SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and 0.5MB model size," arXiv:1602.07360, Feb. 2016.
- [13] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "MobileNets: Efficient convolutional neural networks for mobile vision applications," arXiv:1704.04861, Apr. 2017.
- [14] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, Jun. 2016, pp. 770-778. doi: 10.1109/CVPR.2016.90.
- [15] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in Advances in Neural Information Processing Systems 25 (NIPS 2012), Lake Tahoe, NV, USA, Dec. 2012, pp. 1097-1105.
- [16] S. Han, H. Mao, and W. J. Dally, "Deep Compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding," in 4th International Conference on Learning Representations (ICLR), San Juan, Puerto Rico, May 2016.
- [17] A. Maciá-Lillo, S. Barrachina, G. Fabregat, and M. F. Dolz, "Optimizing convolutions for deep learning inference on ARM Cortex-M processors," IEEE Internet of Things Journal, vol. 11, no. 15, pp. 26203-26219, Aug. 2024, doi: 10.1109/IIOT.2024.3393546.
- [18] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "MobileNets: Efficient convolutional neural networks for mobile vision applications," arXiv:1704.04861, Apr. 2017.
- [19] D. Qin, C. Lechner, M. Delakis, M. Fornoni, S. Luo, F. Yang, W. Wang, C. Banbury, C. Ye, B. Akin, V. Aggarwal, T. Zhu, D. Moro, and A. Howard, "MobileNetV4: Universal models for the mobile ecosystem," arXiv:2404.10518, Apr. 2024.
- [20] M. Rusci, A. Capotondi, and L. Benini, "Memory-driven mixed low precision quantization for enabling deep network inference on micro-controllers," arXiv:1905.13082, May 2019.

TABLE I
COMPARISON OF LIGHTWEIGHT CNN ARCHITECTURES FOR EMBEDDED IOT DEPLOYMENT.

Model	Year	Parameters	Top-1 Acc. (ImageNet)	Memory (in MBs)	MCU Suitability
SqueezeNet [12]	2016	1.25M	57.5	4.8	Low
SqueezeNet (Deep Compressed) [12]	2016	1.25M	57.5	0.47	Fairly High [17]
1.0 MobileNet-224 (FP32) [18]	2017	4.2M	70.6	16.8	Low
MobileNet-v1 (Mixed Precision) [20]	2019	4.2M	68	2	Fairly High
MNv4-Conv-S (FP32) [19]	2024	3.8M	73.8	15.2	Low